

NAGESWAR REDDY BELLAM

GENERATIVE AI & LLM SPECIALIST | AI SOLUTION ARCHITECT | NLP & MULTIMODAL AI EXPERT

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AI Engineering leader with 9+ years of experience in architecting and implementing advanced Machine Learning and Generative AI solutions for finance, legal, healthcare, and automotive sectors. Expertise in LLMs, RAG, and multimodal AI pipelines enables delivery of scalable, production-grade AI solutions that solve high-impact business challenges. Adept at leading end-to-end AI initiatives, from data strategy and model development to deployment and optimization.

Core Competencies

- Large Language Models (LLMs) Fine-Tuning & Deployment
- Natural Language Processing (NLP) & Text Analytics
- Generative AI & RAG (Retrieval Augmented Generation)
- Vector Databases & Semantic Search
- Computer Vision & Deep Learning for Image Classification
- Prompt Engineering & Multimodal AI Pipelines
- ML Model Development & Deployment
- AI Integration with Cloud Platforms
- Automation of Business Processes using AI
- Transformers & Modern ML Frameworks
- AI-as-a-Service (AlaaS)
- Knowledge Graphs / Ontology Engineering
- Multimodal AI

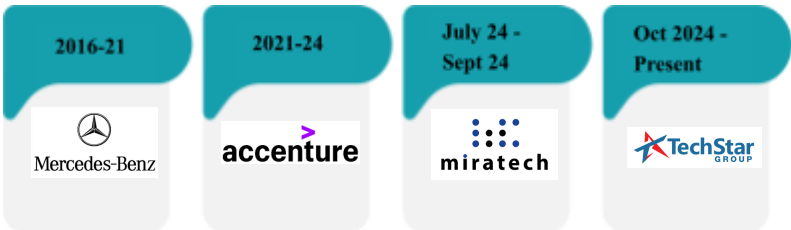
Technical Skills

- Programming & Frameworks:** Python, PyTorch, TensorFlow, Scikit-Learn, HuggingFace, LangChain, LangGraph
- ML/NLP Tools:** NLTK, SpaCy, BERT, Llama2, Gemini Pro, PEFT/LoRA
- Databases & Cloud:** Milvus, Vector DBs, BigQuery, Firestore, AWS, GCP, Vertex AI, Azure AI
- Development Tools:** Docker, K8, CI/CD, GIT, FastAPI, Gradio
- OS:** Linux, Windows

Profile Summary

- Strategic AI Leadership:** 9+ years of experience designing and deploying enterprise-grade Machine Learning and Generative AI solutions across finance, legal, healthcare, and automotive domains, enabling business transformation and process automation.
- Generative AI & LLM Expertise:** Deep expertise in Large Language Models (LLMs), Retrieval Augmented Generation (RAG), and multimodal AI pipelines for solving complex business problems, information extraction, and predictive analytics.
- End-to-End AI Solution Delivery:** Proven track record in leading AI initiatives across the full lifecycle from data preprocessing, model development, and fine-tuning to deployment, optimization, and continuous learning delivering scalable, production-ready systems.
- Innovation & Business Impact:** Skilled at translating strategic business requirements into AI-driven solutions, including virtual assistants, document abstraction pipelines, and predictive systems, generating measurable efficiency gains and actionable insights.
- AI Solution Scalability & Governance:** Expertise in designing AI systems with production-grade scalability, model governance, and compliance considerations, ensuring enterprise-ready deployment and risk mitigation.
- Cross-Domain AI Innovation:** Proven ability to adapt cutting-edge AI/ML technologies across multiple domains, including legal, finance, healthcare, and automotive, driving innovation and measurable business outcomes.

Career Timeline



Work Experience

- AI Engineer | Technical Lead | Techstar Group, CBRE | Sept 2024 – Present**
- Building ontology framework for abstraction of structured data from diverse document types.
 - Extracting and classifying fields from documents with structure and consistency using RAG-based pipelines.
 - Parsing PDFs, storing raw text in Vector DB (Milvus), and performing semantic search to send relevant chunks to GPT-4o for abstraction.
 - Collaborating with SMEs to fine-tune prompts and improve extraction accuracy over conventional ML-NLP solutions.
 - Designing pipelines to automate entity extraction, reducing manual intervention and development time.
 - Leading initiatives to integrate multimodal AI solutions for self-help video generation using Gemini Pro Vision model.
 - Implementing production-grade LLM services with scalable architecture for enterprise use.
 - Monitoring model performance and continuously improving abstraction pipelines using feedback loops.
 - Driving AI innovation for new business processes, identifying opportunities for automation and optimization.
 - Mentoring and guiding junior AI engineers in LLM fine-tuning, prompt engineering, and pipeline development.



Achievements



Led the design and deployment of LLM-based AI solutions, improving document processing efficiency by 12% across finance and legal operations.



Implemented AI-driven underwriting and financial analysis solutions, enabling faster risk assessment and narrative generation.



Education



**M.Tech.,
Manufacturing
Technology** – NIT
Trichy, India
(2016)



**B.Tech.,
Mechanical
Engineering** –
JNTU
Anantapuram,
India (2013)

AI Engineer | Miratech India Pvt. Ltd. | July 2024 – Sept 2024

- Developed healthcare chatbot using DialogFlow CX, integrating with BigQuery and Firestore for customer data management.
- Implemented intent recognition & created webhooks for dynamic information extraction.
- Leveraged Vertex AI and Gemini LLM for intent classification and summarization.
- Automated customer query resolution and improved virtual assistant efficiency.

AI Engineering Senior Analyst | Accenture, Hyderabad | July 2021 – July 2024

- Designed and implemented AI solutions using LLMs for Legal and Finance domains.
- Developed text preprocessing pipelines and fine-tuned transformer-based models for sequence classification.
- Built RAG-based solutions leveraging LangChain, vector stores, and prompt engineering.
- Collaborated with business stakeholders to improve ML model accuracy and solution adoption.
- Developed information extraction & text classification solutions for production deployment.

Machine Learning Engineer | Mercedes Benz R&D Pvt. Ltd. | July 2016 – June 2021

- Developed and deployed ML/DL solutions for automotive digital development and testing.
- Fine-tuned CNN models (VGG16, ZF Net) for automated identification of corroded parts.
- Built custom CNN architectures achieving significantly faster forward pass times.
- Conducted POCs for AI/ML applications in vehicle design, development, and maintenance.



Projects Undertaken

Virtual Agent for Differentiated Customer Experience

- Developing a virtual agent to deliver personalized customer support using DialogFlow CX
- Classifying customer queries by intent and recommending resolution steps via LLM-based automation.
- Enhancing customer service efficiency by intelligently routing queries to the appropriate channels.
- Designing a multimodal pipeline using Gemini Pro Vision to generate self-help videos from product manuals, reducing human intervention.

Entity Abstraction from Documents

- Implemented Retrieval Augmented Generation (RAG) pipelines to extract structured information from diverse document categories.
- Built end-to-end pipeline including text extraction, preprocessing, chunking, vector embeddings, semantic search, and LangChain integration.
- Extracted 30+ fields (names, dates, amounts, receipt numbers) with improved accuracy and 25% faster development time compared to traditional ML-NLP methods.
- Collaborated with SMEs to optimize prompts and improve model performance.

Spreading and Underwriting Automation

- Automated entity extraction from financial documents (balance sheets, income statements) and generated narrative insights for underwriters.
- Processed and queried structured financial tables for LLM consumption.
- Leveraged one-shot and few-shot prompting to improve accuracy and consistency of narrative generation.

Feedback Tag AI Insight

- Built an AI system to analyze user feedback and tag voice assistant conversations.
- Used larger LLMs to generate multiple completions and fine-tuned smaller models for cost-effective, high-performance feedback tagging.
- Implemented an open-source LLM-based solution to reduce operational costs while maintaining high accuracy.

Intelligent Email Advisor

- Developed NLP solution for extracting key information and identifying lines of business for a leading insurance client.
- Fine-tuned BERT models for intent classification and LoB identification, enabling automated email processing and workflow improvements.

Identification of Corroded Parts

- Built deep learning models to automatically identify corroded automotive parts from underbody images captured during testing and service.

- Fine-tuned VGG16 and ZF Net architectures; developed a custom CNN architecture achieving 7x faster forward pass than VGG16 and 2x faster than ZF Net.
- Improved classification accuracy and efficiency, contributing to predictive maintenance and vehicle quality assurance.



Personal Details

Address	: Hyderabad, India
Languages Known	: English, Telugu, Hindi
Date of Birth	: 07-June-1992